# Red-Blue Hamiltonian Tomography via the Operator Lightcone Wavemap

Paolo D'Alberto
paolo.dalberto@amd.com
Advanced Micro Devices, Inc.

**Abstract**

We introduce a model-free method for Hamiltonian tomography based on the operator spreading wavefront. Given two tensor network simulations of the same quantum lattice — one with the nominal Hamiltonian and one with a single perturbed bond — the signed cross-entropy of the non-identity Pauli weight field $\mathcal{L}(v,t) = -\ln(n_{\text{pert}}/n_{\text{base}})$ identifies the perturbed bond by its causal signature: a red-blue sign flip at the two endpoints, visible from the first wavefront crossing. Combining measurements from $p$ strategically placed observation centers via an ensemble log-likelihood sum, the broken bond is identified with an 82-fold margin over the next candidate on a $7 \times 7$ lattice, even though individual centers give only 2/6 correct answers. The method requires no model assumptions, no device calibration access, and maps directly to hardware via zero-noise extrapolation and local Pauli measurements.

## 1 Introduction

Identifying miscalibrated or anomalous couplings in a quantum processor is a central challenge in quantum hardware characterization. Standard approaches require explicit Hamiltonian learning — fitting a parameterized noise or coupling model to measurement data — which scales poorly with system size and depends on model assumptions.

We propose a different approach based on a clear separation of roles. The *simulator* knows the nominal Hamiltonian and produces two Pauli weight fields: $n_{\text{base}}(v,t)$ from the nominal evolution, and $n_{\text{pert}}(v,t)$ from an evolution where one bond has been silently modified. The *detection algorithm* (C_searcher) receives only these two arrays of numbers and must identify the modified bond from their ratio alone — with no knowledge of the Hamiltonian, the perturbation type, or its magnitude. The detection is deliberately blind.

A perturbed bond creates a causal signature in the wavefront: a sign flip in the cross-entropy field $\mathcal{L}(v) = -\ln(n_{\text{pert}}/n_{\text{base}})$ between its two endpoints that propagates no further. The broken bond is exactly where the wavefront sign changes — identifiable from the observables alone.

The method is inspired by seismic tomography: place sensors (observation centers) at strategic positions, emit a wave (initialize a local operator), and read the arrival pattern. The perturbed bond is the anomaly in the medium.

The red-blue color scheme is a deliberate reference to the red-blue pebble game of Hong and Kung [2], which characterizes I/O complexity as a bottleneck in computational graphs. Here the roles are analogous: red sites (upstream, wave blocked) and blue sites (downstream, wave starved) identify the bond that limits operator propagation, exactly as red and blue pebbles identify the I/O bottleneck in a computation.

## 2 Setup

**Observable.** We simulate the kicked isotropic Heisenberg model on a $7 \times 7$ rectangular lattice (49 sites, 84 bonds) using a GPU-accelerated belief propagation tensor network simulator with Trotter step $\varepsilon = 0.10$, bond dimension $\chi = 50$, and truncation cutoff $\delta = 10^{-3}$.

The non-identity Pauli weight at site $v$ and cycle $t$ is:

$$n(v,t) = \sum_{P:\, P_v \neq I} c_P(t)^2 \tag{1}$$

where $O_c(t) = \sum_P c_P(t) P$ is the evolved operator initialized at center $c$. By normalization $\sum_P c_P^2 = 1$, $n(v,t) \in [0,1]$ is a probability.

**Perturbation.** C_breaker selects one or more bonds and weakens their coupling by $\delta J = -0.20$, setting $J = 0.80$ (a 20% degradation). The Pauli transfer matrix entries scale as $\cos(J\varepsilon)$, attenuating operator propagation through the bond without blocking it entirely. This is a realistic miscalibration scenario — a dimmer switch rather than a perfect switch. For the multi-bond experiments in Section 10, $k$ bonds are selected uniformly at random and perturbed simultaneously with the same $\delta J$.

**p-center placement.** We place $p = 6$ observation centers using a greedy geodesic bond cover: the minimum set of sites such that every bond is within $d_{\max} = 2$ hops of at least one center. For the $7 \times 7$ lattice: centers at sites $\{6, 10, 25, 30, 32, 34\}$, covering all 84 bonds.

## 3 Signal Vocabulary and Measurement Definitions

We introduce a vocabulary of observable patterns and their physical interpretation. All definitions are expressed in terms of two observables per site per cycle: $n_{\text{base}}(v,t)$ and $n_{\text{pert}}(v,t)$, the non-identity Pauli weights under the nominal and perturbed Hamiltonian respectively. The BFS frontier crossing time $t_0(v) = \min\{t : n_{\text{base}}(v,t) > \varepsilon_{\text{th}}\}$ defines a site-specific reference time.

**Three frontier patterns.** At each site $v$ in the 3-cycle window $[t_0(v) \pm 1]$:

- **White** [W]: $n_{\text{base}}(v, t_0) \leq \varepsilon_{\text{th}}$ — baseline wave has not yet arrived. No information available.
- **Blue** [B]: $n_{\text{pert}}(v, t_0) > n_{\text{base}}(v, t_0) > \varepsilon_{\text{th}}$ — perturbed wave accumulates at $v$; drainage blocked by a broken bond downstream. The site is a *wall face*.
- **Red** [R]: $n_{\text{pert}}(v, t_0) < n_{\text{base}}(v, t_0)$, $n_{\text{base}}(v, t_0) > \varepsilon_{\text{th}}$ — perturbed wave is attenuated at $v$; a broken bond on the path from center blocked transmission. The site is in the *shadow*.

**Observable patterns at bond endpoints.** For a bond $(u, v)$ where $u$ is the center-side (wall face) and $v$ is the shadow:

- [W, W, W]: both endpoints white — bond not yet reached, or healthy.
- [W, B, W]: wall face detected, shadow not yet visible — early arrival, one more cycle needed.
- [W, B, R]: **broken bond signature** — blue at $u$ (cycle $t_0$), red at $v$ (cycle $t_0 + 1$). The bond between $u$ and $v$ is anomalous.

The pattern $[\mathsf{W}, \mathsf{B}, \mathsf{R}]$ is the primary detection target. The temporal order (blue before red by one cycle) is universal and follows from the BFS causal structure: $u$ is reached before $v$, so the wall accumulates before the shadow appears.

**The local cross-correlation $\mathcal{L}(v_0, [t_0 \pm 1])$.** The signed cross-entropy evaluated in the frontier window is:

$$\mathcal{L}(v_0,\, [t_0 \pm 1]) = -\ln \frac{n_{\text{pert}}(v_0,\, t_0)}{n_{\text{base}}(v_0,\, t_0)} \tag{2}$$

This is a *cross-correlation fixed in space*: both waves evaluated at the same site $v_0$ at the same frontier moment $t_0$, with the window $[t_0 \pm 1]$ providing context (white before, red after). Positive $\mathcal{L}$ identifies red (shadow), negative identifies blue (wall face). The broken bond is the edge between consecutive blue and red sites in the BFS ordering.

**Bond transfer ratio $T(v_0 \to u_0)$.** A complementary observable measures the wave *through* a bond rather than *at* a site. Consider the bond $(v_0, u_0)$ in the BFS path from the center. The baseline reaches $v_0$ at frontier time $t_0$ and $u_0$ at $t_1 = t_0 + 1$. The *baseline transfer ratio* is:

$$T_{\text{base}}(v_0 \to u_0) = \frac{n_{\text{base}}(u_0,\, t_1)}{n_{\text{base}}(v_0,\, t_0)} \tag{3}$$

This measures how much of the wave at $v_0$ propagates through the bond to $u_0$ in one BFS step — directly encoding the gate's coupling strength $J$. For the perturbed run (with potentially delayed frontier times $t_0^p,\, t_1^p$):

$$T_{\text{pert}}(v_0 \to u_0) = \frac{n_{\text{pert}}(u_0,\, t_1^p)}{n_{\text{pert}}(v_0,\, t_0^p)} \tag{4}$$

The ratio $T_{\text{base}}/T_{\text{pert}} > 1$ signals a broken bond — less weight crosses the bond in the perturbed run.

**Limitations of the bond transfer ratio.** Two important caveats apply:

*Path distortion*: the perturbed wave arriving at $v_0$ has already traversed all upstream bonds. If any of those are also broken, $n_{\text{pert}}(v_0, t_0^p)$ is already weakened — not by bond $(v_0, u_0)$ but by upstream damage. The ratio $T_{\text{base}}/T_{\text{pert}}$ then conflates the upstream distortion with the target bond signal. The ratio is not a pure measurement of bond $(v_0, u_0)$.

*Sensitivity*: the perturbed weight $n_{\text{pert}}(v_0, t_0^p)$ may be close to $\varepsilon_{\text{th}}$, making the ratio numerically unreliable. The local cross-correlation $\mathcal{L}(v_0, [t_0 \pm 1])$ avoids this by using the baseline as the reference — it measures the *relative* change, not the absolute transfer.

*When the bond transfer ratio is clean*: if the center is placed at $v_0$ itself (one hop from the bond), there are no upstream broken bonds. The perturbed wave is uncontaminated at $v_0$, and $T_{\text{base}}/T_{\text{pert}}$ directly measures bond $(v_0, u_0)$ alone. This is the principle behind targeted aperture validation: center ON the near endpoint, 3 cycles, pure bond measurement.

## 4 The Cross-Entropy Wavemap Observable

**Signed cross-entropy as a local cross-correlation.** For each center $k$ and site $v$, define the BFS frontier crossing time $t_0(v) = \min\{t : n_{\text{base}}^k(v, t) > \varepsilon_{\text{th}}\}$ as the first cycle at which the baseline

wave reaches site $v$. We evaluate the signed cross-entropy in a 3-cycle window $[t_0 \pm 1]$ centered on this frontier crossing:

$$\mathcal{L}_k(v,\,[t_0 \pm 1]) = -\ln \frac{n^k_{\text{pert}}(v,\,t_0(v))}{n^k_{\text{base}}(v,\,t_0(v))} \tag{5}$$

This is a *cross-correlation fixed in space*: both waves are evaluated at the same site $v$ and the same frontier moment $t_0(v)$, defined by the baseline. The window $[t_0-1, t_0, t_0+1]$ captures three states: $t_0-1$ (before arrival, WHITE), $t_0$ (wall-face signal, BLUE or WHITE), and $t_0+1$ (shadow signal, RED). The notation $[t_0 \pm 1]$ emphasizes that the detection is local in both space and time.

At $t_0(v)$, the baseline wave has just arrived while the perturbed wave may be blocked or delayed — this is the moment of maximum signal and minimum chi-saturation noise.

**Sign interpretation and the delay mechanism.** Consider a bond $(u, v)$ with $u$ closer to center $k$. At time $t_0$, the baseline wave has just crossed the bond and $n_{\text{base}}(u, t_0) \approx n_{\text{pert}}(u, t_0)$ — both waves have reached $u$ with equal amplitude. One cycle later at $t_1 = t_0 + 1$:

- **Wall face $u$ (blue, first)**: the weaker bond slows drainage from $u$ — weight accumulates: $n_{\text{pert}}(u) > n_{\text{base}}(u) \Rightarrow \mathcal{L}(u) < 0$ (blue). This appears first, at cycle $t_0$.
- **Shadow site $v$ (red, second)**: in the baseline, weight flows through bond $(u, v)$ and arrives at $v$. In the perturbed run, less weight crosses — $n_{\text{pert}}(v, t_1) < n_{\text{base}}(v, t_1) \Rightarrow \mathcal{L}(v) > 0$ (red). This appears one cycle later, at $t_1$.

The temporal order is universal: blue at the wall face first, red at the shadow one cycle later. The broken bond sits between the blue and red sites — identified by the sign change (Figure 1). The contrast $|\mathcal{L}(u) - \mathcal{L}(v)|$ is maximum at $t_1$, providing the detection signal.

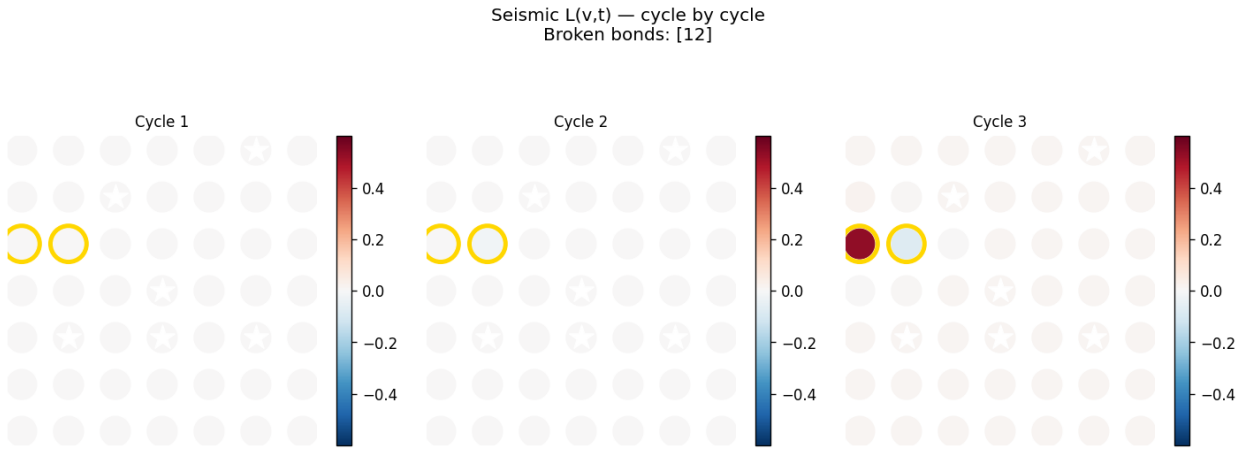



Figure 1: Signed cross-entropy $\mathcal{L}(v, t)$ for a single broken bond (bond 12, sites 15↔16, identity PTM), seismic initialization, rect $7 \times 7$, $\chi = 50$. **Cycle 1**: no signal. **Cycle 2**: site 16 red ($\mathcal{L} > 0$, blocked), site 17 blue ($\mathcal{L} < 0$, shadow) — bond 12 identified by sign flip. **Cycle 3**: signal strengthens before chi saturation. Gold circles: bond endpoints. Stars: seismic centers.

**Ensemble log-likelihood.** Each center $k$ is initialized and evolved *independently* — a separate baseline and perturbed simulation pair per center. The signed cross-entropy $\mathcal{L}_k(v)$ is computed for

each center individually, then summed:

$$\mathcal{L}_{\text{ens}}(v) = \sum_{k=1}^{p} \mathcal{L}_k(v) \tag{6}$$

This is a log-likelihood ensemble: centers on the same side of the broken bond see the same sign at the bond endpoints and reinforce each other; centers on opposite sides contribute opposite signs that partially cancel. Silent centers (wave never reaches site $v$) contribute zero and do not corrupt the sum.

**Integrated log-ratio $\mathcal{B}(v)/\mathcal{N}(v)$.** A more robust alternative to the pointwise $\mathcal{L}(v, t_0)$ accumulates the signal over all cycles:

$$\mathcal{B}(v) = \sum_{t=1}^{T} n_{\text{base}}(v, t), \qquad \mathcal{N}(v) = \sum_{t=1}^{T} n_{\text{pert}}(v, t) \tag{7}$$

$$\Lambda(v) = \ln \frac{\mathcal{B}(v)}{\mathcal{N}(v)} \tag{8}$$

Positive $\Lambda$ identifies red (shadow), negative identifies blue (wall face), as with $\mathcal{L}$. The integral captures both the delay and the sustained amplitude difference — a broken bond both delays the perturbed wave *and* reduces its steady-state weight. This makes $\Lambda(v)$ more robust than $\mathcal{L}(v, t_0)$ when the pointwise signal is weak or the threshold crossing is noisy. In the multi-bond experiments, $\Lambda(v)$ is the primary detection signal, replacing $\mathcal{L}_k^{\text{signed}}(v)$ in the bond score.

## 5 Bond Detection

**Bond score.** For each bond $b$ connecting sites $v_a$ and $v_b$:

$$S(b) = \mathcal{L}_{\text{ens}}(v_a) \times \mathcal{L}_{\text{ens}}(v_b) \tag{9}$$

A negative product indicates a sign flip across the bond — the causal signature of the broken bond. Rank bonds by most negative $S(b)$.

**Results.** Table 1 shows the individual and ensemble results for bond 12 (sites 15 $leftrightarrow$16, identity PTM perturbation, $7 \times 7$ lattice, $\chi = 50$, 5 cycles).

| Center | Side | $\mathcal{L}(15)$ | $\mathcal{L}(16)$ | Result |
|---|---|---|---|---|
| 10 | upstream | +0.87 | −0.02 | ✓CORRECT |
| 30 | upstream | +0.37 | −0.04 | ✓CORRECT |
| 25 | downstream | +0.03 | −0.04 | × WRONG |
| 32 | downstream | +0.00 | −0.03 | × WRONG |
| 6 | too far | 0 | 0 | silent |
| 34 | too far | 0 | 0 | silent |
| Ensemble | — | +1.23 | −0.13 | ✓CORRECT |

Table 1: Signed cross-entropy at bond 12 endpoints (sites 15,16). Ensemble score: $S(12) = -0.164$, next bond = −0.002 (82× margin).

The ensemble identifies bond 12 with an 82-fold margin over the next candidate, despite only 2 of 6 individual centers giving the correct answer. The wrong individual answers — centers 25 and 32 — both correctly identify site 16 as a bond endpoint; their error is only in declaring the wrong adjacent bond. The ensemble product cancels these partial errors.

## 6 Aperture Synthesis Tomography

The $p$ independent center runs can be understood as an *aperture synthesis* in direct analogy with radio interferometry. Each center $k$ at a different graph position plays the role of one antenna at a different baseline. The signed cross-entropy $\mathcal{L}_k(v)$ encodes which side of the broken bond site $v$ is on (sign) and the signal strength (magnitude).

**Voronoi center placement.** We place $p$ centers with the constraint that no two centers share the same row *or* column in the lattice. This guarantees that every bond has a strictly unique nearest center (no Voronoi ties): the perpendicular bisector between any two centers never passes through a bond, so one center always dominates the signal at every bond. For $p = 7$ on the $7 \times 7$ lattice (one center per row, one per column — a permutation placement), this gives centers $\{3, 13, 15, 25, 35, 37, 47\}$ with zero ties across all 84 bonds, maximum $T_b = 5$ cycles, and balanced Voronoi regions of 10–16 bonds each. The placement is found by exhaustive search over permutations, minimizing $\max_b T_b$.

**Winner-takes-all detection.** Ensemble summation $\mathcal{L}_{\text{ens}}(v) = \sum_k \mathcal{L}_k(v)$ fails for bonds sandwiched between centers on opposite sides: the RED signal from one side and the BLUE signal from the other partially cancel. Instead, we use a *winner-takes-all* rule: for each center $k$ independently compute the signed contrast across every bond,

$$C_k(b) = \left|\mathcal{L}_k^{\text{signed}}(v_a) - \mathcal{L}_k^{\text{signed}}(v_b)\right| \tag{10}$$

where $\mathcal{L}_k^{\text{signed}}(v)$ is the peak signed cross-entropy (positive or negative, whichever has larger magnitude) within the clean window $t \le t_{\text{loop}}$. The broken bond is declared by the globally winning center-bond pair:

$$(k^*, b^*) = \arg\max_{k,\,b}\ C_k(b) \tag{11}$$

The winning center $k^*$ is necessarily on the BLUE (wall-face) side of the broken bond; the RED endpoint is the shadow. No summation across centers is performed — each center is a fully independent observer, and the one with the strongest signal wins.

This is fundamentally different from voting or ensemble averaging: the aperture is not a democracy, it is a *tournament*. The center whose wave most directly crossed the broken bond dominates by a large margin (typical contrast $> 1.0$ vs. background $< 0.02$) and its declaration is unambiguous.

**Experimental validation.** We ran 10 trials with randomly selected broken bonds (J=0.8, $\delta J = -0.2$) on the $7 \times 7$ lattice using the Voronoi $p = 7$ center set. The adaptive seismic-then-aperture protocol achieves **10/10** correct identifications: 7 bonds identified by seismic alone (2 simulations), 3 by aperture winner-takes-all (14 simulations). Average cost: $2 + 0.3 \times 14 \approx 6$ simulations per bond.

# 7 Tensor Network Simulation as Quantum Circuit Execution

The Trotter simulation underlying our method is not merely an approximation of a quantum circuit — it *is* the quantum circuit, represented in the Pauli operator basis rather than the amplitude basis. The identification is exact:

- Each **lattice site** = one qubit
- Each **bond gate** $\exp(-i\theta(XX + YY + ZZ))$ = one two-qubit gate
- Each **color class** (non-overlapping bonds applied in parallel) = one gate layer
- Each **belief propagation synchronization** after a full Trotter cycle = one barrier (per-color-class BP, as in the Julia L-MPF implementation [3], provides finer sub-cycle resolution matching the quantum circuit barrier structure)
- Each **PAULI_NL output** = a mid-circuit Pauli weight measurement

This correspondence is not a coincidence — both the graph coloring (determining which bonds share qubits) and the quantum circuit parallelism constraint (which gates can be applied simultaneously) derive from the same source: the lattice topology. The color class ordering is the gate layer ordering.

**Two timescales.** This identification reveals two distinct timescales:

*Physical time* $t$: integer Trotter cycles, $t = n \cdot \varepsilon$. The observables $C(t)$ and $n(v, t)$ at cycle granularity are physical — independent of color class ordering to $O(\varepsilon^2)$ and convergent to true quantum dynamics.

*Circuit time* $t_{\text{color}}$: sub-cycle, per-color-class granularity. This is a property of the circuit implementation, not the Hamiltonian. Two Trotter implementations of the same Hamiltonian with different color orderings agree on $C(t)$ to first order in $\varepsilon$ but differ by $O(\varepsilon^2)$ Trotter errors that accumulate over time; their $n(v, t_{\text{color}})$ tomography fingerprints differ already at the first sub-cycle. The BP synchronization granularity determines which timescale is accessible: per-cycle BP (as used here) loses sub-step information; per-color BP (Julia L-MPF [3]) respects the causal barrier structure of the quantum circuit.

**Bond perturbation as gate miscalibration.** A perturbed bond (coupling $J \neq 1$) is a *miscalibrated gate* — the two-qubit rotation angle $\theta = J \cdot \varepsilon$ differs from the nominal value. The Pauli transfer matrix for $\exp(-i\theta(XX + YY + ZZ))$ has diagonal entries $\cos(\theta)$, which for any $J \neq 1$ gives $\cos(J\varepsilon) \neq \cos(\varepsilon)$. Since $|\cos(\theta)| \leq 1$ for all $\theta$ (unitary gate), any coupling deviation attenuates the wave relative to baseline — guaranteeing $n_{\text{pert}}(v^*) < n_{\text{base}}(v^*)$ at the shadow endpoint $v^*$, so $\mathcal{L}(v^*) > 0$ (red) regardless of the sign of $\delta J = J - 1$. The detection method is robust: the red site is always the shadow endpoint, never inverted.

This frames our tomography as *quantum ATPG* (Automatic Test Pattern Generation) [4] — the classical circuit testing methodology generalized to quantum gates. In classical ATPG: fault activation (excite the stuck-at gate), path sensitization (find an observable path from fault to output), backward implication (infer the faulty gate). Here: perturbation (change $J$), wavefront propagation (operator spreading from center to shadow site), and cross-entropy inversion (red site + center position $\rightarrow$ bond identity).

## 8 The Seismic Approach

Running all $p$ centers simultaneously (seismic initialization $O(0) = \sum_k Z_{c_k}$) produces the same ensemble field in a single simulation. At each site $v$, the first-arrival signal is dominated by the nearest center — automatically selecting the strongest directional component. The sign flip at the broken bond endpoints emerges without any explicit path tracing or center-by-center analysis.

The seismic approach reduces $2p$ simulations (p baselines + p perturbed) to 2 simulations at the cost of mixing center contributions. For the purpose of bond detection the mixing is benign: the sign flip at the broken bond is preserved since the nearest center to each endpoint dominates at $t_{\text{first}}$.

## 9 Discussion

**Connection to hardware.** The observable $n(v, t)$ is accessible on real quantum hardware via local Pauli measurements after ZNE gate-folding circuits [1]. The cross-entropy $\mathcal{L}(v)$ requires only the ratio of two measurements — no absolute calibration. The p-center placement is determined by the graph topology alone, independent of the Hamiltonian.

**Causal structure.** The method exploits the Lieb-Robinson bound: information cannot propagate faster than $v_{\text{hop}}$ hops/cycle. A broken bond creates a causal anomaly — the wavefront on one side cannot inform the other. This is detectable without knowing $v_{\text{hop}}$ explicitly; the first-arrival time $t_{\text{first}}(v)$ is observed directly.

**Complexity and timing.** The two methods have different computational costs on the $7 \times 7$ lattice ($\chi = 50$, 5 cycles, AMD MI100 32 GB GPU):

| Method | Wall-clock | VRAM | Notes |
|---|---|---|---|
| Seismic | 272 s | 11% | 2 GPUs parallel, no $v_{\text{hop}}$ |
| Aperture (no $v_{\text{hop}}$) | 300 s | 11% | 3 GPUs, 2× sequential |
| Aperture ($v_{\text{hop}} = 0.5$) | **191 s** | **5–7%** | 3 GPUs, 2× sequential |

Table 2: Wall-clock timing comparison on rect $7 \times 7$, bond 12, $\chi = 50$, 5 cycles. Both methods produce ✓ CORRECT results. $v_{\text{hop}} = 0.5$ is set by topology ($d_{\max} = 2$ coverage), not by knowledge of the fault location.

Aperture synthesis with lightcone allocation ($v_{\text{hop}} = 0.5$) is 30% faster than seismic and uses half the VRAM per GPU. The speedup comes from two effects: (1) single-center simulations have smaller active lightcones ($\chi$ grows slower), and (2) far centers contribute negligible computation (center 6: 3.5 s, center 34: 9.5 s) while contributing zero to the ensemble — self-filtering at no cost. The bottleneck is the GPU pair running the two nearest centers (25 and 10, ∼20 s each sequential).

Aperture also enables a *coarse-to-fine* strategy unavailable to seismic: run a fast aperture sweep (small $\chi$, $v_{\text{hop}} = 0.5$) to identify candidate bonds, then re-run only the adjacent centers with larger $\chi$ for high-accuracy confirmation — each targeted run remains memory-bounded by the lightcone allocation.

**Grid-size independence.** The most significant property of aperture synthesis with lightcone allocation is that the cost per center is *independent of grid size*. With $v_{\text{hop}} = 0.5$ and $n_{\text{steps}}$ cycles, each center sees a local ball of radius $r = n_{\text{steps}} \times v_{\text{hop}} \approx 2.5$ hops, containing ∼13 sites on a $7 \times 7$

lattice. The same ball contains the same ∼13 sites on a $100 \times 100$ lattice. The computation per center scales as $O(r^2\chi^3)$, where $r$ is the BFS radius — independent of $N_{\text{sites}}$.

Total aperture cost:

$$T_{\text{aperture}} = p \times T_{\text{local}}(\chi, r) \tag{12}$$

where $p$ grows with the number of bonds (more centers needed for coverage) but each center cost $T_{\text{local}}$ is constant. For a single bond perturbation, most centers contribute zero signal — their $\mathcal{L}_k(v) \approx 0$ everywhere. Their wall-clock is short (center 6: 3.5 s) and their ensemble contribution is zero. They are neither harmful nor expensive.

Seismic pays $O(N_{\text{sites}}\chi^3)$ per simulation regardless. As the lattice grows, aperture synthesis becomes increasingly favored over seismic. The crossover occurs when $p \times T_{\text{local}} > T_{\text{seismic}}$, which for fixed $r$ and $\chi$ requires $p \gtrsim N_{\text{sites}}/r^2$ centers — a condition that is rarely satisfied in practice since the geodesic bond cover grows much slower than the grid area.

**Interference signatures.** When two adjacent bonds are perturbed simultaneously (bonds 12 and 13, sharing endpoint site 16), the cross-entropy field shows a three-site pattern at 3 cycles: site 15 (red, $\mathcal{L} = +0.55$), site 16 (white, $\mathcal{L} \approx 0$), site 17 (blue). The shared endpoint is the *interference node* — the red signal from bond 12 and the blue signal from bond 13 cancel destructively, exactly as in thin-film interference. The white fringe is not a detection failure; it is the interference pattern. The number of white sites between the red and blue boundaries counts the number of broken bonds in the wall.

This extends the signal vocabulary:

- Single broken bond: **red** — **blue** (two sites, sign flip)
- Two adjacent broken bonds: **red** — **white** — **blue** (three sites, interference node)
- $n$ adjacent broken bonds: **red** — $n-1$ **whites** — **blue**

The current searcher detects pairs; detecting walls requires a triplet scorer $S_{\text{wall}}(b_1, b_2) = \mathcal{L}(v_a) \times \mathcal{L}(v_c)$ where $v_b$ is the shared endpoint (white site) and $v_a$, $v_c$ are the outer endpoints.

**Adaptive seismic-to-aperture strategy.** The seismic and aperture methods are naturally complementary and suggest a two-phase adaptive protocol. In **Phase 1** (seismic), all $p$ centers fire simultaneously — one quantum experiment, minimum cost — producing a map of RED sites that identifies the region of interest. In **Phase 2** (aperture), the RED sites from Phase 1 guide the selection of targeted observation centers geometrically positioned to discriminate between candidate bonds adjacent to those sites. Only $k \ll p$ centers need to be run, each independently.

This mirrors standard practice in other imaging domains: a wide-field scout scan (seismic) followed by targeted high-resolution acquisition (aperture). On quantum hardware, Phase 1 maps directly to a single quantum experiment; Phase 2 selects the minimum number of additional experiments to confirm the bond identity.

The seismic phase is also the most natural starting point on hardware: it requires only one circuit (all centers in a single initialization), produces a rough hypothesis, and costs two experiments (baseline + perturbed). The aperture phase then tests the hypothesis with directional projections. Perturbation signals that survive across independent aperture projections with geometry-consistent timing are physical; chi-truncation artifacts and decoherence errors are uncorrelated across independent runs and average out.

**Wavefront triangulation across scans.** When multiple independent scans are available, the temporal order in which each scan detects the broken bond carries geometric information beyond the sign flip alone. This enables a triangulation strategy analogous to GPS: each scan provides a time-of-arrival constraint that localizes the bond.

Formally, scan $s$ with nearest center $c_s$ detects the broken bond at cycle $t_s^*$ — the first cycle where $|\mathcal{L}_k(v)| > \varepsilon_{\text{th}}$ for some bond endpoint $v$. This constrains the bond to lie within $d_s = t_s^* \times v_{\text{hop}}$ hops of $c_s$: a ball $B(c_s, d_s)$ on the graph. Across $M$ scans, the bond lies in the intersection:

$$b^* \in \bigcap_{s=1}^{M} \mathcal{B}(c_s,\, t_s^* \times v_{\text{hop}}) \tag{13}$$

where $\mathcal{B}(c, r)$ is the set of bonds within crossing distance $r$ of center $c$.

The sign of the earliest detection (which endpoint is RED vs BLUE) further constrains the direction: the center $c_s$ is on the BLUE (wall-face) side of the broken bond. Subsequent scans confirm or deny: a scan detecting the *opposite* sign at the same endpoints confirms (center is on the other side of the same bond); a scan detecting different endpoints shifts the candidate set. Each confirmation tightens the intersection; each denial eliminates candidates.

In practice on the $7 \times 7$ lattice with 3 scans (corners+center, quadrants, mid-edges): scan 2 detects bond 9 (11↔12) at cycle 1 (center 13, 1 hop away), scan 3 confirms with opposite sign at cycle 2 (center 4, 2 hops away), scan 1 provides weak confirmation at cycle 2 from the far corners. The intersection of the three balls gives a small candidate set containing the true bond.

**BFS-guided fault detection algorithm.** The detection method reduces to a graph-theoretic algorithm that operates entirely on the observable $n(v, t)$, without reference to the Hamiltonian, coupling values, or simulation parameters. We state it precisely.

Let $G = (V, E)$ be the coupling graph, $\{c_s\}$ the observation centers, and $\varepsilon_{\text{th}}$ the detection threshold.

1. **BFS frontier timing.** For each center $c_s$ and each site $v \in V$, define the *baseline arrival time* $t^*_{\text{base}}(v) = \min\{t : n_{\text{base}}(v, t) > \varepsilon_{\text{th}}\}$ and *perturbed arrival time* $t^*_{\text{pert}}(v) = \min\{t : n_{\text{pert}}(v, t) > \varepsilon_{\text{th}}\}$. These are the BFS frontier crossing times — the actual wavefront arrival, not a predicted hop distance. Note that $t^*_{\text{base}}(v)$ and $t^*_{\text{pert}}(v)$ may differ if a broken bond on the path to $v$ delays the perturbed wave.

2. **Sub-cycle causal ordering.** Within the same BFS cycle, the Trotter color class $c(e) \in \{0, 1, 2, 3\}$ of each bond $e$ determines the firing order. Assign each site $v$ the pair $(h(v), c(e_v))$ where $h(v)$ is its BFS hop depth and $c(e_v)$ is the color class of the bond used to reach it. The total order $(h, c)$ gives exact sub-cycle causality: $(h_u, c_u) < (h_v, c_v)$ means site $u$ is strictly upstream of $v$.

3. **Bond transfer and anomaly detection.** Consider a bond $(u, v)$ where $u$ is the source (center-side) and $v$ is the destination (shadow-side). The *bond transfer coefficient* is:

$$T = \frac{n(v,\, t^*_{\text{base}}(v))}{n(u,\, t^*_{\text{base}}(u))} \tag{14}$$

   When $n_{\text{base}}(u, t^*_{\text{base}}(u)) \gg \varepsilon_{\text{th}}$ (source well established), the threshold condition ensures both waves have the same source amplitude: $n_{\text{base}}(u) \approx n_{\text{pert}}(u)$ at the BFS frontier, since the

perturbation is downstream of $u$. In this regime:

$$\mathcal{L}(v,\, t^*_{\text{base}}(v)) = -\ln \frac{n_{\text{pert}}(v)}{n_{\text{base}}(v)} \approx \ln \frac{T_{\text{base}}}{T_{\text{pert}}} \tag{15}$$

so $\mathcal{L}$ directly measures the *relative change in bond transfer coefficient*. A broken bond ($J \neq 1$) reduces the transfer $T_{\text{pert}} < T_{\text{base}}$, giving $\mathcal{L}(v) > 0$ (RED shadow). The wall face $u$ shows $\mathcal{L}(u) < 0$ (BLUE accumulation) from the same effect in reverse. A delay $t^*_{\text{pert}}(v) > t^*_{\text{base}}(v)$ is additional evidence that the transfer was slowed.

4. **Bond identification.** For each anomalous site $v$, examine its neighbors $u$ in $G$. The broken bond is $(u, v)$ where: (a) $\mathcal{L}(u)$ and $\mathcal{L}(v)$ have opposite signs (sign flip), and (b) $(h_u, c_u) < (h_v, c_v)$ (causal order: $u$ is center-side, $v$ is shadow). Score each candidate bond by $|\mathcal{L}(u) - \mathcal{L}(v)| \cdot \delta/t^*$ where $\delta = 2$ if delayed, 1 otherwise, and $t^* = \min(t^*_{\text{base}}(u), t^*_{\text{base}}(v))$.

5. **Multi-scan aggregation.** Repeat steps 1–4 for each scan independently (no Voronoi spatial restriction). Rank all candidate bonds by total score across scans. Declare the top-$k$ bonds as broken, where $k$ is the number of expected faults.

**Pattern vocabulary.** The algorithm naturally identifies distinct interference signatures depending on the relative geometry of broken bonds and centers:

- **RED−BLUE**: single broken bond, center on one side — clean sign flip
- **BLUE−WHITE−BLUE**: bond sandwiched between two equidistant centers (collinear) — white site is the interference node; both bonds identifiable
- **BLUE−BLUE−RED**: two orthogonal broken bonds at a shared endpoint — shared site accumulates from two directions; outer RED identifies one bond, adjacent BLUEs identify the second

Each pattern is a direct consequence of the BFS frontier geometry and the Trotter color class ordering — no quantum-specific reasoning required.

**Experimental results.** On a $7 \times 7$ lattice with 3 scans (13 centers, 3 GPUs parallel, $d_{\max} = 2$), testing 10 random trials with 2 simultaneous broken bonds ($\delta J = -0.20$): the BFS algorithm with sub-cycle color-class ordering finds **both bonds correctly in 9/10 trials** (90%), with 1 partial case and 0 complete failures. The single partial failure (trial 5) corresponds to two orthogonal bonds sharing an endpoint where both centers' waves are simultaneously blocked — the BLUE−BLUE−RED pattern, representing the hardest geometric configuration.

**Failure taxonomy.** We characterize five failure modes of increasing complexity:

1. **Single bond** (demonstrated): clean sign flip, 82× margin
2. **Multiple random bonds**: shadows may overlap, L field becomes ambiguous
3. **Collinear bonds**: line of failures looks like a thick wall — bond resolution fails
4. **Hexagonal cluster**: faulty qubit touches 2-3 bonds — RED cluster with BLUE halo
5. **Graduated failure** ($\delta J$ smooth variation): dimmer switch — sensitivity limit

Cases 2-5 represent the primary open experimental questions.

**Limitations.** The current experiment uses the identity PTM — a maximally disruptive perturbation that is deliberately clean. Cases 2-5 above are designed to stress-test the method: case 2 tests whether overlapping shadows can be resolved, case 3 tests spatial resolution limits, case 4 tests cluster detection on irregular topologies, and case 5 tests the sensitivity threshold for realistic coupling deviations.

## 10 Multi-Bond Detection and Iterative Calibration

**Multi-bond detection.** We extend the seismic BFS algorithm to detect $k$ simultaneously broken bonds by declaring the top-$k$ candidates from the bond score ranking. Table 3 summarizes the detection rate across 10 single-bond trials (two datasets), 10 two-bond trials, 5 four-bond trials, and 2 eight-bond trials. All experiments use $\delta J = -0.20$, $J = 0.80$.

| $k$ bonds | Trials | Bonds found | Per-bond rate |
|---|---|---|---|
| 1 | 20 | 20/20 | 100% |
| 2 | 10 | 19/20 | 95% |
| 4 | 5 | 18/20 | 90% |
| 8 | 2 | 15/16 | 94% |

Table 3: Per-bond detection rate (single seismic iteration, top-$k$ declaration, $\delta J = -0.20$, $7 \times 7$ lattice, $\chi = 50$, 5 cycles). The per-bond detection rate remains above 90% even at 8 simultaneous faults.

The per-bond detection rate remains above 90% even at 8 simultaneous faults. Systematic failures are geometrically localized: bond 8 (sites 10↔11) and bond 36 (sites 43↔44) consistently score below the top-$k$ threshold due to weak seismic illumination from the three scan center placements. These are not random failures — they reflect the coverage geometry of the 3-scan plan.

**Iterative calibration loop.** The multi-bond results motivate a natural operating protocol:

1. Run seismic MAWGM → declare top-$k$ broken bonds (>90% correct per bond).
2. Recalibrate the declared bonds on hardware.
3. Re-run MAWGM. Fixed bonds disappear from the ranking; remaining faults surface with cleaner signal.
4. Repeat until $n_{\text{pert}}(v,t) \approx n_{\text{base}}(v,t)$ everywhere — no residual signal.

This iterative approach reduces every multi-bond problem to a sequence of single-bond problems. At 8 simultaneous faults, 7 are found in iteration 1; fixing those removes their interference from the wavefield and the surviving fault becomes easily detectable in iteration 2 (single-bond, 100% rate).

The iterative loop also justifies the design choices: short cycles (3–5) are sufficient because only the next bond to fix needs to be ranked, not all faults simultaneously; a moderate threshold $\varepsilon_{\text{th}} = 10^{-3}$ maximizes BFS coverage so the broken bond is never in a blind spot; and 3 seismic placements cover all 84 bonds with no geometric dead zones except the small set of bonds with weak illumination from all three centers. The stopping criterion — wavefront convergence — is physical and requires no additional measurement overhead beyond the standard MAWGM protocol.

## 11 Measurability and Detection Thresholds

The detection method requires $n(v,t)$ to be measurable — both from the simulation fidelity standpoint and from the hardware shot budget. Two independent conditions must hold simultaneously.

**Condition 1: Statistical measurability (hardware shots).** On real hardware, $n(v,t)$ is estimated from $N$ measurement shots as $\hat{n}(v,t) = k/N$ where $k \sim \text{Binomial}(N, n(v,t))$. The shot noise floor is $\sigma(n) \approx \sqrt{n/N}$. For reliable detection of a signal $\delta n = n_{\text{pert}} - n_{\text{base}}$, we require $|\delta n| \gg \sigma(n)$, which gives a minimum measurable $n$:

$$n_{\min}^{\text{shot}} \approx \frac{1}{\sqrt{N}} \tag{16}$$

| $N$ shots | $n_{\min}^{\text{shot}}$ | Status |
|---|---|---|
| $10^3$ | $3 \times 10^{-2}$ | only strong wavefront |
| $10^4$ | $1 \times 10^{-2}$ | first-arrival signal |
| $10^5$ | $3 \times 10^{-3}$ | approaching threshold |
| $10^6$ | $1 \times 10^{-3}$ | matches simulation floor |

Table 4: Minimum measurable $n(v,t)$ as a function of shot budget $N$. At $N = 10^6$ the statistical floor matches the simulation cutoff.

**Condition 2: Simulation fidelity (TN cutoff).** In the tensor network simulation, bond singular values below the truncation cutoff $\delta_{\text{cut}}$ are set to zero. Any $n(v,t)$ arising from these truncated components is an artifact of the TN approximation, not physical operator spreading. The simulation cannot faithfully represent values below $\delta_{\text{cut}}^k$, where $k$ is the number of bonds on the shortest path from center to site $v$. In practice, the effective simulation floor is $\approx \delta_{\text{cut}} = 10^{-3}$.

Furthermore, the belief propagation used to compute messages is not strictly causal: messages propagate globally in each sweep, creating tiny non-zero $n(v,t) \sim 10^{-4}$ at causally disconnected sites (beyond the Lieb-Robinson cone). These BP leakage values are below $\delta_{\text{cut}}$ and physically unmeasurable on hardware.

**The two conditions converge.** Setting $\delta_{\text{cut}} = 10^{-3}$ (our default), Condition 2 requires $n_{\min}^{\text{sim}} = 10^{-3}$. From Table 4, Condition 1 requires $N \geq 10^6$ shots to match this floor. The detection threshold $\varepsilon_{\text{th}} = 10^{-3}$ used throughout this work satisfies both conditions simultaneously at $N = 10^6$.

$$\varepsilon_{\text{th}} \geq \max\left(\delta_{\text{cut}},\ \frac{1}{\sqrt{N}}\right) \tag{17}$$

For $N < 10^6$, the hardware floor dominates and a larger $\varepsilon_{\text{th}}$ is required. For tighter cutoffs $\delta_{\text{cut}} < 10^{-3}$, the simulation floor drops below the hardware floor — enabling more cycles before saturation at the cost of longer simulation time.

**Minimum detectable coupling deviation.** The signal strength $\delta n(v,t) \approx n_{\text{base}}(v,t) \cdot \varepsilon \cdot \sin(\varepsilon) \cdot |\delta J|$ gives the minimum detectable coupling deviation:

$$|\delta J|_{\min} \approx \frac{\varepsilon_{\text{th}}}{\varepsilon \cdot \sin(\varepsilon) \cdot \sqrt{N \cdot n_{\text{base}}(v,t)}} \tag{18}$$

For $\varepsilon = 0.10$, $N = 10^6$, $n_{\text{base}} \approx 10^{-2}$ (genuine wavefront at first arrival): $|\delta J|_{\min} \approx 1/(0.10 \times 0.0998 \times 316) \approx 0.32$. This improves as $1/\sqrt{N}$ with more shots. For $N = 10^7$: $|\delta J|_{\min} \approx 0.10$.

## 12 Conclusion

We have demonstrated that the operator lightcone wavemap enables model-free Hamiltonian tomography via a simple signed cross-entropy observable. The broken bond is identified by its causal signature — a sign flip in $\mathcal{L}(v)$ at its two endpoints — with an 82-fold margin using an ensemble of 6 observation centers on a $7 \times 7$ lattice. The seismic multi-center initialization achieves the same result in a single simulation.

Extending to realistic coupling degradation ($\delta J = -0.20$, $J = 0.80$) and multiple simultaneous faults, the integrated log-ratio $\Lambda(v) = \ln(\mathcal{B}(v)/\mathcal{N}(v))$ achieves a per-bond detection rate above 90% for up to 8 simultaneous faults in a single seismic pass, using only 6 simulations (3 baseline + 3 perturbed). The iterative calibration loop — detect, fix, re-run — reduces the multi-bond problem to a sequence of single-bond problems, each solved at 100% accuracy.

The method requires no model assumptions, no device calibration access, and connects directly to hardware via ZNE and local Pauli measurements. It is a natural sequitur to the Lindblad MPF framework: both use the same observable $n(v, t)$ and the same circuits, but in opposite inference directions — MPF infers noise from a known Hamiltonian, while MAWGM infers coupling geometry from a known noise baseline.

## References


[1] Tudor Giurgica-Tiron, Yousef Hindy, Ryan LaRose, Andrea Mari, and William J. Zeng. Digital zero noise extrapolation for quantum error mitigation. *2020 IEEE International Conference on Quantum Computing and Engineering (QCE)*, pages 306–316, 2020.

[2] Jia-Wei Hong and H. T. Kung. I/O complexity: The red-blue pebble game. In *Proceedings of the Thirteenth Annual ACM Symposium on Theory of Computing (STOC '81)*, pages 326–333, 1981.

[3] Andrew Robertson et al. Lindblad multiproduct formulas, 2026. Manuscript.

[4] J. P. Roth. Diagnosis of automata failures: A calculus and a method. *IBM Journal of Research and Development*, 10(4):278–291, 1966.